

DEVCOM SC-BOTAA 22-01 REQUEST FOR PROJECT PROPOSAL (RPP)

Area of Interest: Soldier Sustainment Directorate, Aerial Delivery Division
Prototype: Contested Aerial Delivery Development

Objective

This Request for Project Proposal (RPP) is issued to request pre-proposals (white papers) in accordance with Section III, B, 1 of the U.S. Army Combat Capabilities Development Command Soldier Center (DEVCOM SC) Broad Other Transaction Authority Announcement (BOTAA) 22-01 competitive announcement and is being issued for planning purposes only. It is not to be construed as a commitment by the Government to award an agreement, nor will the Government pay for the information solicited. Product Manager Force Sustainment Systems (PdM FSS) is seeking information on the capabilities and willingness of private entities (non-profit and commercial) to develop a preliminary Contested Aerial Delivery Development (CADD) design that will support a prototype project, as defined on page 5 of the BOTAA.

Background

In a contested, Anti-aircraft / Area Denial (A2/AD) environment, traditional forms of aerial delivery are not possible due to the presence of both kinetic and electronic defenses. Advances are required in aerial delivery vehicles to increase their operable deployment range from target, which puts aircraft out of harm's way. Additionally, advances in sensor technology are required to allow these aerial delivery vehicles to operate nominally in the presence of electronic warfare.

Requirements

The purpose of this RPP is to request pre-proposals to conduct system development, testing, and evaluation of a preliminary design for a CADD capability. The preliminary design may be based on reusing, modifying, or enhancing existing designs.

The pre-proposals shall address the full requirements of this RFPP for a technology that incorporates:

Develop an Airdroppable Air Vehicle Prototype

1. Payload Capability

- The system must securely transport a payload of at least 250 pounds, with the capability to accommodate payloads of varying shapes and dimensions up to 48 x 96 x 108 inches.
- The air vehicle must include mechanisms to mitigate vibration and shock during flight to ensure payload integrity, meeting a shock tolerance of 8g for up to 5 milliseconds, as demonstrated in independent testing or by using a surrogate known platform.
- The payload compartment must provide quick-release latching mechanisms for efficient unloading
- Demonstrate successful delivery of payloads with no significant damage across 5 airdrop trials under operational conditions.

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2. Range Requirement

- The air vehicle must achieve a minimum range of 150 nautical miles, with verification through flight tests using full payload weight under standard atmospheric conditions (15°C, sea level).
- Performance must be maintained at operational altitudes of 5,000 to 15,000 feet MSL, with no degradation in range beyond 25% due to wind or turbulence.
- Include fail-safe systems to ensure controlled descent or return-to-base if the range capability is exceeded.

3. Airdrop Compatibility

- The system must integrate with military-standard aerial delivery systems, including Type V platforms and CDS bundles, and comply with MIL-STD-1791 for airdrop operations.
- The air vehicle must support release velocities between 90 and 130 knots indicated airspeed (KIAS).
- Include provisions for manual and automatic release modes, with release success rates meeting 92% reliability with 90% confidence for any system that will be fielded

4. Post-Airdrop Stability

- The system must autonomously stabilize within 90 seconds of release, maintaining a heading deviation of no more than ± 45 degrees during stabilization.
- Demonstrate sustained flight performance under wind conditions of up to 18 knots steady-state and 25 knots gusting.
- Implement onboard sensors and control algorithms to ensure yaw, pitch, and roll stability with deviations not exceeding 15 degrees per second during cruise.
- Validate stability performance through at least 30 airdrop tests, achieving a minimum success rate of 95% stability metrics.

Develop a Sensor Kit Prototypes for Navigation in GPS Denied Environments

1. Accuracy Performance

- The navigation system must achieve a positional accuracy of ≤ 150 meters Circular Error Probable (CEP) when compared to truth data from an independent reference system.
- Accuracy must be validated in diverse operational scenarios, desert, forests, and open terrain, through a multiple test runs per scenario.
- The system must maintain accuracy for up to 5 hours of continuous operation without external GPS input.
- Provide periodic calibration capabilities to ensure sustained accuracy, with recalibration intervals of 30 minutes or less.

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2. Environmental Robustness

- The sensor kit must operate within a temperature range of 0°C to +40°C and at humidity levels of up to 95% non-condensing.
- Fielded system performance must not degrade in environments with dust (ISO 12103-1 A2), heavy rain (5 mm/min), or electromagnetic interference (EMI) at levels of up to 50 V/m.

3. Integration and Modularity

- The sensor kit must provide modular components, supporting integration with existing navigation systems using open-standard interfaces (e.g., RS-232, CAN, or Ethernet).
- Enable plug-and-play functionality for seamless swapping of components, with integration and configuration times of under 2 hours.
- Include detailed interface control documents (ICDs) and software development kits (SDKs) to support custom integrations.
- Validate integration with at least three unique platforms during testing.

4. Initialization and Operation

- The sensor kit must achieve full initialization within 60 seconds of power-on, supporting dynamic startup conditions (e.g., while in motion).
- Provide a continuous navigation solution during startup, with positional drift not exceeding 3% of distance traveled.
- Validate operational readiness through at least 10 dynamic initialization tests across multiple aerial platforms.

Develop Associated Tool Prototypes

1. Mission Planning Software

- Develop a mission planning tool capable of generating optimized flight paths based on user-defined waypoints, payload characteristics, and environmental factors (e.g., wind speed, altitude).
- Include a user-friendly interface with drag-and-drop waypoint configuration, real-time flight simulations, and mission validation checks.
- Integrate terrain data and no-fly zone overlays to ensure compliance with operational and regulatory constraints.

2. Communication Tools

- Design a communication system supporting secure, real-time data exchange between the air vehicle and ground control station over encrypted channels compliant with AES-256 standards.
- Implement dynamic frequency selection and adaptive modulation to ensure reliable communication in contested environments.

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- Include tools for monitoring communication link status, latency, and bandwidth utilization during operations.

3. Command & Control Tools

- Develop a command and control (C2) interface to enable operators to monitor vehicle status (e.g., position, speed, fuel/power levels) and issue commands (e.g., rerouting, altitude adjustments) in real time.
- Support autonomous and manual control modes, with seamless transitions between modes during flight.
- Include logging capabilities to capture C2 inputs and system responses for post-mission analysis.

4. Associated Engineering

- Provide hardware and software engineering improvements for prototypes which do not result in increasing the current performance envelope
- Provide hardware and software engineering to support enhancements for hardware and software which has become obsolete during the course of the effort

Deliverables

1. Integrated Master Schedule. Within ten (10) business days of the start of work, the contractor shall provide an Integrated Master Schedule (MS Project or PDF format).
2. Contractor Spend Plan. Within ten (10) business days of the start of work, the contractor shall provide a spend plan that estimates the delivery date and amount of each invoice that the contractor anticipates to deliver over the life of the task/contract. The spend plan is only an estimate and the contractor shall revise that estimate as work progresses.
3. Monthly Status Report. The monthly report shall be delivered not more than the 15th following each month. The report shall be in Microsoft Word® electronic format. The monthly report shall include:
 - a. Activities completed
 - b. Meetings and briefings attended
 - c. Project status
 - d. Planned activities
 - e. Cost for the reporting period
4. The contractor shall deliver software documentation packages in conjunction with executable capability packages. Documentation packages shall include, for each software application/plugin:
 - a. Software User's Manual (SUM)
 - b. Installable Application and Execution Code
 - c. Source Code and Data Files

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- d. Software Release Engineering Documentation. Once per year, the Government will require the following documentation for software released to the end-user:
 - e. Software Development Plan (SDP) (DI-IPSC-81427A)
 - f. Software Requirements Specifications (SRS) (DI-IPSC-81433A) / Interface Requirements Specification (IRS) (DI-IPSC-81434A)
 - g. Requirements Traceability Matrix (RTM)
 - h. Software Design Document (SDD) (DI-IPSC-81435A) / Interface Design Document (IDD) (DI-IPSC-81436A)
 - i. Software Test Report (STR) (DI-IPSC-81440A)
 - j. Software Test Plan (STP) (DI-IPSC-81438A)
 - k. Software Test Description (STD) (DI-IPSC-81439A)
 - l. Software Product Specification (SPS) (DI-IPSC-81441A)
 - m. Software Support Environment User Guide (SSEUG)
 - n. Software Version Description (SVD) (DI-IPSC-81442A)
5. The contractor shall deliver hardware documentation including:
- a. 2D Technical drawings for TDP. Includes two-dimensional drawings of designs that have been produced for aerial delivery. Must abide by general Geometric Dimension and Tolerancing principles.
 - b. 3D CAD models. Includes three-dimensional computer aided design files models of designs for aerial delivery. Preferable software application used is Solidworks, though others will be accepted
 - c. Engineering design analyses. Documents analyses of designs developed for aerial delivery using general engineering principles such as quality, reliability, robustness, usability, and modularity.
6. Hardware prototypes. Proof-of-concept and functional prototypes to include hardware systems, software, and material solutions. Prototypes, to the greatest practical extent, simulate the final design, aesthetics, materials, and functionality of the intended design. Minor rework or software code modifications will be allowed, provided they do not prevent accurate evaluation of the system or component design. Prototypes shall be utilized in field testing and evaluation when needed. Software code shall meet the needs of stakeholders, with sound software engineering discipline applied to the design, development, and deployment of interoperable products.
7. Test Reports. The contractor shall deliver a test report for each flight test that was supported. The report shall include purpose of test, configuration of hardware software, software log file summary, and analysis of results. The analysis will include a determination of the root cause of any unexpected results as well as recommendations for future tests.

Performance Objectives

Pre-proposals should be limited to ten (10) pages and include a description of the relevant technology including supporting data, the scope of the proposed effort including a high-level Work Breakdown Structure (WBS), and a description of the proposer's research, development, manufacturing, past performance, or other special qualifications. The pre-proposal should indicate the TRL of the proposed design per the TRL definitions found in the attached

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Technology Readiness Assessment Guidebook. The pre-proposal shall also include an anticipated cost for the efforts described in the pre-proposal. Pre-proposals shall be submitted to both of the following email address: Christine Halamoutis, Agreements Specialist at christine.m.halamoutis.civ@army.mil and Elizabeth Nordell, Agreements Officer at elizabeth.m.nordell.civ@army.mil.

Brochures or other descriptions of general organizational or individual capabilities will not be accepted as a pre-proposal. All pre-proposals will be assigned an identification number, and an email will acknowledge receipt of a pre-proposal. Usually, the Project Manager of the submitting organization should receive a decision letter or email regarding the pre-proposal within 60-90 days of submission.

Vendor(s) that may ultimately receive a Request for Full Proposal (RFFP) shall provide supporting information within the full proposal on the following topics in addition to what is specified in DEVCOM SC BOTAA Section III, B, 2:

Project Management

Analysis of any opportunities and/or benefits associated with the vendor's proposal that would reduce cost, schedule, and performance risks in Phase II.

Pricing

Firm fixed pricing for each deliverable, in the vendor's own format/template along with any supporting documentation so that the Agreement Officer is able to make a fair and reasonable determination.

Substantiating information on costs for each deliverable.

Administration

Any questions regarding this Request for Project Proposal (RPP) shall be submitted via email to Christine Halamoutis, Agreements Specialist at christine.m.halamoutis.civ@army.mil and Elizabeth Nordell, Agreements Officer at elizabeth.m.nordell.civ@army.mil. The Government will retain comments and information received in response to this RPP. Proprietary information should be identified as such. Do not use Government security classification markings. All written responses must be received via email no later 10:00 a.m. EST **on March 20, 2025** with a confirmation email received in response to the submission. Responses should be sent by e-mail to the two points of contact above with the subject line of organization and Request for Project Proposal (RPP) title. Material that is advertisement only in nature is not desired. Respondents may review the DEVCOM SC BOTAA at <https://sam.gov/opp/126eaf3a78d847f69b7c94b606877f92/view> for additional areas of interest and/or pre-proposal submission instructions and representative POCs. All submissions must include a completed Non-Traditional Defense Contractor (NTDC) Certification, attached to posting.